An Improved Fuzzy Region Competition-Based Framework for the Multiphase Segmentation of SAR Images

Shiyu Luo, *Member, IEEE*, Kamal Sarabandi, *Fellow, IEEE*, Ling Tong, *Member, IEEE*, and Sen Guo

***Abstract*— The objective of this article is to investigate a multiphase segmentation framework for synthetic aperture radar (SAR) images, which is proposed based on the idea of the fuzzy region competition-based method. The fuzzy region competition-based framework is highly efficient and can attain good segmentation performances for conventional images. The framework is achieved based on its convexity, which not only ensures the existence of a globally optimized solution but also enables the convex optimization theory-based solving algorithms that are feasible. However, the constraint conditions of the framework that guarantee this convexity probably cannot be satisfied in the segmentation of images corrupted with strong noise. Therefore, applying this method to an SAR image probably produces an unsatisfactory segmentation result. To address this problem, we propose an improved fuzzy region competition-based framework in terms of the hierarchical strategy, such that the framework is always convex during the iterative calculation. The proposed framework inherits the advantages of the fuzzy region competition-based method, as well as that it is able to be applied to the segmentation of images with strong noise. Several experiments are then carried out to test and verify the performance and the robustness of the proposed framework. It demonstrates that the proposed segmentation framework can be applied to various types of SAR images and achieves satisfactory segmentation results.**

***Index Terms*— Active contour model, fuzzy function, hierarchical strategy, region competition, synthetic aperture radar (SAR) image segmentation.**

I. Introduction

IMAGE segmentation plays an important role in image processing and computer vision. The aim of segmentation is to partition an image into multiple pairwise disjoint regions with smooth boundaries and homogenous characteristics. Various kinds of segmentation methods have been developed in recent years, such as Markov random field (MRF)-based methods [1], K-means-based methods [2], fuzzy C-means-based methods [3], and convolutional network-based methods [4]. These methods have also been successfully applied to synthetic aperture radar (SAR) images [5]. An expectation–maximization algorithm is another one of commonly used segmentation methods. This kind of methods achieves image segmentation by maximizing *a posteriori* probability in terms of a Bayesian estimation principle. The key point of such methods is to solve a minimum energy functional.

The emergence of "snake" model using gradient information promotes the rapid development of such type of methods [6]. Based on this model, extensive studies have been developed. For example, Cohen [7] proposed a Balloon model that improves the capability of capturing edges; Xu and Prince [8] developed a gradient vector flow (GVF) model to solve the problem of the poor capability of detecting groove edges. In these models, contour lines are denoted by a closed parametric curve. Such an explicit representation goes against the topology change, such as split and uniting. To overcome this drawback, the level set proposed by Osher and Sethian [9] was introduced to image segmentation, which converts explicit representation to implicit representation by embedding low-dimensional curves into high-dimensional surfaces. Based on the "snake" model and the curve evolution theory, Caselles *et al*. [10] proposed a geodesic active contour (GAC) model built on a variational method and a level set method. This model is geometric intrinsic because its evolution function does not involve any other parameters in relation to the geometric structure of the curve except for unit normal vector and curvature. Aimed at images without distinct edges, the models based on region information were developed. The most representative model is the active contours without edges model proposed by Chan and Vese (CV model) [11]. This model uses average information instead of edge information to decide segmentation boundaries. For the need of segmentation of images polluted by noise, antinoise ability is another problem to focus on. The commonly used model is a region competition model proposed by Zhu and Yuille [12], which is built based on the Bayesian and minimum description length criterion. The Bayesian error in this method is estimated by parametric probability distributions, where the statistical distributions are chosen in terms of different images, for example, Gaussian distribution is commonly used for optical image segmentation [12]–[14]. Due to strongly

Manuscript received June 17, 2019; revised September 19, 2019; accepted October 18, 2019. Date of publication November 14, 2019; date of current version March 25, 2020. This work was supported by the Natural Science Foundation of China under Grant 41571333 and Grant 41901277. *(Corresponding author: Shiyu Luo.)*

S. Luo and L. Tong are with the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: shiyuluo@uestc.edu.cn).

K. Sarabandi is with the Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, MI 48109 USA.

S. Guo is with the School of Economics and Management, North China Electric Power University, Beijing 102206, China.

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2949742

0196-2892 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

multiplicative noise presented in SAR images, to ensure high segmentation accuracy, a more precise fitting distribution function is required in terms of different SAR images. For example, the gamma distribution is applied to SAR images at low resolutions [15]–[17], the Weibull distribution is used in SAR images with homogenous regions [18], the Fisher distribution is suited to the SAR image segmentation of urban areas [19], and so on. The level set approaches can attain good performances for most of the cases, whereas they have an intrinsic convergence limitation [20]. Because neither the involved energy functional nor the involved set of characteristic functions is convex, which leads to two shortcomings. One is low efficiency. A locally convex solving method based on the variational principle and the gradient descent flow theorem for the global minimization is used to solve level set functional. This solving method requires reinitialization of level sets to distance functions in each numerical iteration and is computationally inefficient. The other is the existence of local minima. This affects the algorithm's converging to the global minimum and therefore produces a poor segmentation result.

To solve the above-mentioned defects, two-phase segmentation methods using fuzzy membership functions (FMFs) based on the idea of region competition, referred to as fuzzy region competition-based methods, have been proposed [20]–[22]. It proves that the energy functional associated with FMFs is convex, such that globally minimum exists and an efficient solving algorithm built on the globally convex optimization theory can be used [23], [24]. Multiphase segmentation is a more challenging task. The most difficult is to find an effective and distinct representation of each region. Following the same line of the two-phase fuzzy region competition-based methods and the multiphase segmentation methods based on level sets [25]–[27], some multiphase segmentation methods associated with FMFs have been developed [28], [29]. To ensure the convexity of the involved functional in these methods, FMFs are required to belong to a convex set and are subjected to two convex conditions. However, we found that the two convex conditions probably cannot be satisfied simultaneously during the numerical process when applying the methods to an SAR image. This makes the involved functional nonconvex and thus leads to a failure in segmentation. This disadvantage manifests more with the increasing number of segmented regions. Consequently, such methods are very competitive and can attain satisfactory segmentation of images contaminated with no or weak noise, whereas they are not suited to SAR images.

To circumvent the above-mentioned problem, we propose a hierarchical strategy-based multiphase segmentation framework based on the idea of the two-phase fuzzy region competition-based model [28]. The overview of the proposed method and the contributions are summarized as follows. First, we rewrite the fuzzy region competition-based framework in terms of hierarchical strategy, and then, we define the segmentation criterion in terms of the constraint conditions with respect to FMFs. Second, starting with the first FMF from the fuzzy membership family, the involved functional is minimized by fixing the rest of FMFs. Since the involved functional and the FMF are both convex, the fast dual-projection algorithm developed based on the convex optimization theory is thus employed to implement segmentation. Proceeding to the following FMFs, we then give the solutions for the involved functional. Finally, we discuss the estimation of used characteristics parameters in functional and the termination criterion. In the proposed method, the two convex conditions are always met simultaneously during the process of calculation. Consequently, such a promising segmentation framework, namely the fuzzy region competition-based method, is successfully introduced to the segmentation of SAR images. The proposed framework not only inherits the advantages of the fuzzy region competition-based method but also has the good capability of noise immunity. By utilizing the appropriate image characteristics, the proposed segmentation framework can be applied to various types of SAR images and can attain segmentation of an arbitrary number of regions.

The remainder of this article is organized as follows. In Section II, the basic model of a Bayesian framework is briefly introduced, and the fuzzy competition region-based method and its disadvantages are discussed. Section III elaborates the proposed method. Section IV describes the experimental verification. Finally, the conclusion is given in Section V.

II. Related Works

A. Basic Model

The image segmentation problem in terms of the Bayesian framework is formulated as follows: assuming that an image $I(x) : \Omega \to \mathbb{R}$ consists of N regions, where Ω is the image domain which is a bounded open subset of $\mathbb{R}^2$. The family $\{\Omega_i\}_{i=1}^N$ is pairwise disjoint and meets $\bigcup_{i=1}^N \Omega_i = \Omega$. The segmentation problem is achieved by maximizing *a posteriori* probability as follows:

$$\begin{aligned}\{\Omega_i\}_{i=1}^N &= \arg\max_{\{\Omega_i \subset \Omega\}} \prod_{i=1}^N p(\Omega_i | I(x)) \\ &= \arg\max_{\{\Omega_i \subset \Omega\}} \prod_{i=1}^N p(\Omega_i) p(I(x)|\Omega_i) \end{aligned} \tag{1}$$

where $\prod_{i=1}^N p(I(x)|\Omega_i)$ is the likelihood of segmentation and $p(\Omega_i)$ is the prior probability. It is noted that the term $p(I(x))$ in (1) is independent on segmentation results and is thus removed.

To implement segmentation, this problem is converted to a minimization problem by taking the negative logarithm of (1) as follows:

$$\{\Omega_i\}_{i=1}^N = \arg\min_{\{\Omega_i \subset \Omega\}} \mathbf{E}\left[\{\Omega_i\}_{i=1}^N\right] \tag{2}$$

where $\mathbf{E}[\{\Omega_i\}_{i=1}^N]$ is referred to energy functional defined by

$$\mathbf{E} = -\sum_{i=1}^N \left\{ \int_{\Omega_i} [\log(p(I(x)|\Omega_i)) + \log p(\Omega_i)] \mathrm{d}x \right\}. \tag{3}$$

In (3), $-\log(p(I(x)|\Omega_i))$ measures the characteristics of an image within each region. In many studies [30]–[32], this term

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

is modeled as statistically parametric models. The commonly used models are gamma distribution for SAR images. For different types of SAR images composed of various scenes, many researchers have introduced and presented extensively statistical models as well as other characteristics information to energy functional [33]. In the region competition-based models, $\boldsymbol{r}_i(x) = -\log(p(I(x)|\Omega_i))$ describes the underlying model of each region $\boldsymbol{\Omega}_i$ in terms of image characteristics, which is referring to as error functions. Furthermore, the length of divide curve denoted by the regularization term $\oint_{\partial\Omega_i} \mathrm{d}l$ is commonly used as the prior probability $-\log p(\boldsymbol{\Omega}_i)$, where l is the contour line. The energy functional in (3) is then formulated as follows:

$$\mathbf{E} = \sum_{i=1}^{N} \int_{\Omega_i} \boldsymbol{r}_i(x)\mathrm{d}x + \sum_{i=1}^{N} \oint_{\partial\Omega_i} \mathrm{d}l. \tag{4}$$

B. Fuzzy Region Competition-Based Models

Segmentation is achieved by minimizing the energy functional. Minimization of energy functional is generally implemented by the utilization of closed plane curves or level sets. The former one has the low capability of topology changes, and the latter one is computationally inefficient. Besides, local minima exist in the minimization of energy functional by using these two methods, leading to a barely satisfactory segmentation performance. The introduction of FMF to energy functional is able to solve the above-mentioned problems. By introducing a convex fuzzy function set, energy functional is proven to be convex as well [20]. Both convexities assure a globally optimal solution and enable efficiently convex optimization tools that are available. Mory and Ardon [20] propose a two-phase segmentation method based on a fuzzy region competition. This method using fuzzy function set is much more efficient and stable than that uses closed plane curves and level sets, and is also easy to implement. Energy functional in (4) associated with an FMF is defined as follows:

$$\mathbf{E} = \int_{\Omega} \boldsymbol{r}_1(x)u(x)\mathrm{d}x + \int_{\Omega} \boldsymbol{r}_2(x)(1-u(x))\mathrm{d}x + \int_{\Omega} |\nabla u(x)|\mathrm{d}x \tag{5}$$

where ∇ is the Hamiltonian operator. $u(x)$ is an FMF belonging to a convex set. This set is simply chosen to be the space of the functions of bounded variations taking their values in [0, 1]. Please note that $1-u(x)$ belongs to this set as well, namely $(1-u(x)) \in [0,1]$. With a bounded convex set, (5) is convex and thus has only global solutions. We refer readers to [20] for its detailed proof. Consequently, the value of $u(x)$ greater or less than 0.5 decides to which region the corresponding point is classified.

Li *et al.* [28] proposed a multiphase segmentation method based on Mory's model. Their model with respect to (4) is defined as follows [28]:

$$\mathbf{E} = \sum_{i=1}^{N} \left\{ \int_{\Omega} \boldsymbol{r}_i(x)u_i(x)\mathrm{d}x + \int_{\Omega} |\nabla u_i(x)|\mathrm{d}x \right\} \tag{6}$$

where $u_i(x)$ is an FMF from the fuzzy membership family. Each FMF $u_i(x)(i = 1, 2, \ldots, N)$ corresponds to a subdomain Ω_i.

Similarly, to guarantee the convexity of energy functional, each FMF needs to belong to a convex set. Using the same simple convex set as in [20], FMFs in (6) need to satisfy

$$\begin{aligned} &\text{(i)} \quad \sum_{i=1}^{N} u_i(x) = 1 \\ &\text{(ii)} \quad u_i(x) \in [0,1], \quad i = 1, 2, \ldots, N. \end{aligned} \tag{7}$$

Equation (6) under these two constraint conditions is proven to be positive and convex, and we refer readers to [28] for the detailed proof.

During the processing of numerical iteration, the constraint condition (ii) in (7) is probably not satisfied, namely $u_i(x) \notin [0,1]$. To address this problem, an approximate numerical solution $\hat{u}_i(x)$ is calculated by using a formula in each iteration as follows:

$$\hat{u}_i(x) = \min\{\max\{u_i(x), 0\}, 1\}. \tag{8}$$

The combination of (7) and (8) ensures the convexity of (6). Consequently, the point is classified into i-region when its corresponding $u_i(x)$ is the greatest among its corresponding fuzzy membership family.

C. Disadvantage of the Multiphase Fuzzy Competition-Based Segmentation Model

Generally, FMFs $u_i(x)$ are initialized randomly under the constraint conditions in (7). For the two-phase segmentation, the constraint conditions in (7) are satisfied automatically since when $\hat{u}_1(x) \in [0,1]$, $u_2(x) = 1 - \hat{u}_1(x)$ always falls into [0,1]. Equation (6) is therefore consistent with (5). In other words, the implementation of the two-phase segmentation of an image using the fuzzy region competition-based model converges unconditionally to a global solution. For a multiphase segmentation ($N \geq 3$), however, the constraint conditions in (7) probably cannot be satisfied. On one hand, it may occur that more than two FMFs are greater than 1 in the iterative process, namely $u_i(x) > 1$, $u_j(x) > 1$, $i \neq j$. By using (8), $\hat{u}_i(x) = \hat{u}_j(x) = 1$, which leads to $\sum_{i=1}^{N} u_i(x) \geq 2$, and thus, the constraint condition (ii) in (7) is not satisfied. In this case, the fuzzy membership family does not belong to a convex set, leading to that the involved energy functional is not convex in the next iterations. On the other hand, it may occur that $u_j(x)$ is not the greatest among $u_i(x)(i = 1, 2, \ldots, N)(i \neq j)$ for any x. For this case, although the image domain is still covered by the union of FMFs, no point will be placed into j-region. This probably results in the incorrect computation in the next iterations. The above-mentioned two cases cause an incorrect multiphase segmentation. In the multiphase segmentation of an image corrupted without or with weak noise, these negative cases can be effectively avoided by setting a very small iteration step in the numerical iteration process. However, these cases are more likely to occur in the multiphase segmentation of an image polluted by strong noise. Furthermore, a very small

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

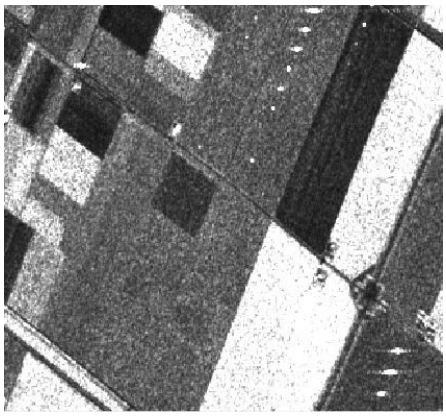


Fig. 1. Backscattering total power image obtained from the test image.

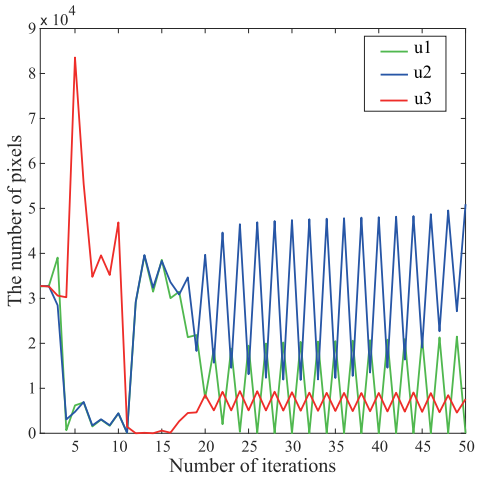


Fig. 2. Number of pixels within each segmented region with respect to Fig. 1 using (6).

iteration step substantially reduces computational efficiency. Consequently, this promising algorithm is theoretically generalized to the segmentation of an arbitrary number of regions, whereas it is hardly applied to the multiphase segmentation of an SAR image in practice.

We use a real SAR image to test the performance of the multiphase fuzzy region competition-based segmentation algorithm. This SAR image contains three homogenous regions (croplands) acquired from Flevoland by AIRSAR at L-band. The image is composed of four polarizations, HH, HV, VV, and VH. The size of the image is 300×325. For the visual examination, the backscattering total power image obtained from the data is shown in Fig. 1. The used characteristics for this SAR image in the energy functional (4) is the statistical mean $\mathbf{V}$ of the coherency matrices $\mathbf{T}$ defined as $\mathbf{T} = \boldsymbol{k}\boldsymbol{k}^{*\mathrm{T}}$, where $\boldsymbol{k} = 1/\sqrt{2}[S_{\mathrm{hh}} + S_{\mathrm{vv}}, S_{\mathrm{hh}} - S_{\mathrm{vv}}, 2S_{\mathrm{hv}}]^{\mathrm{T}}$. S_{hh}, S_{vv}, S_{hv}, and S_{vh} are the four complex elements of a backscattering matrix, where h and v denote the horizontal and vertical polarizations, respectively. Generally, the coherency matrix $\mathbf{T}$ for each point within every region is assumed to follow a complex Wishart distribution as follows [34]:

$$p(I(x)|\Omega_i) = \frac{L^{qL}|\mathbf{T}(x)|^{L-q}\exp\left(-\mathrm{Tr}|L\mathbf{V}_i(x)^{-1}\mathbf{T}(x)|\right)}{\pi^{q(q-1)/2}\Gamma(n)\cdots\Gamma(L-q+1)|\mathbf{V}_i(x)|^{L}} \tag{9}$$

where Γ is the gamma function, Tr is the trace of the matrix, L is the number of looks, q is the order of $\mathbf{T}(x)$, $\mathbf{V}_i(x)$ is the statistical mean of $\mathbf{T}(x)$ within i-region, and $|\ |$ denotes determinant.

Consequently, the error function $\boldsymbol{r}_i(x)$ with respect to region Ω_i in (4) is written as

$$\boldsymbol{r}_i(x) = -\log(p(I(x)|\Omega_i)). \tag{10}$$

The image is segmented into three regions ($N = 3$) by using three FMFs. The segmentation is attained by minimizing the fuzzy region competition-based model in (6) associated with (9) and (10). The number of pixels within each segmented region during every iteration is shown in Fig. 2. We can observe that all the curves are declining in the around 11th iteration in Fig. 2. It indicates that a group of the pixels is not placed to any segmented regions because the sum of the pixels corresponded to the three FMFs is the number of the total pixels, which is supposed to be unchanged. This case can be attributed to the fact that the projection values of more than two FMFs are all equal to 1 after using (8), resulting in the problem that the segmentation rule cannot decide to which region the corresponding points are classified. This fact also leads to that the constraint condition (i) in (7) is not met and the involved energy functional is thus nonconvex. Furthermore, we can observe that no pixels is classified into the region corresponded to $u_3(x)$ in the around 14th iteration because in this case, $u_3(x)$ is not greater than $u_1(x)$ or $u_2(x)$ for any x, namely $\{x|u_3(x) > u_1(x)\bigcap u_3(x) > u_2(x), \forall x\} = \varnothing$. The curves vibrate intensely after then. This leads to an incorrect segmentation eventually, as shown in Fig. 3.

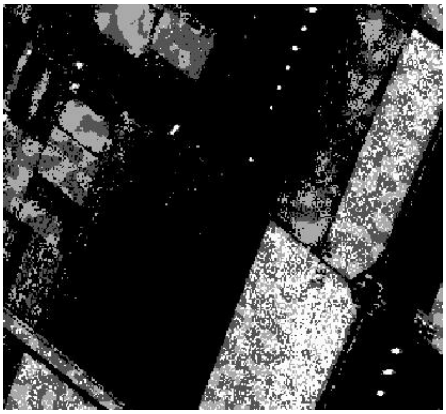


Fig. 3. Segmentation result by the multiphase fuzzy region competition-based segmentation model.

The discussion and the experiment indicate that in order to introduce such a promising framework to the segmentation of SAR images, it is necessary to develop a new framework that not only has the advantages of the fuzzy region competition-based models but also can be applied to images polluted by strong noise.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

III. Proposed Framework

A. Introduction of the Hierarchical Strategy to the Segmentation Framework

Section II indicates that with a bounded convex set of FMFs, the minimization of the involved energy functional converges unconditionally to a global solution. Based on this conclusion, we propose an improved multiphase fuzzy region competition-based segmentation framework in terms of the hierarchical strategy. In the proposed framework, the minimization of the involved energy functional is calculated layer by layer in every iteration. In each layer, only one FMF is used in the calculation, while the rest of FMFs are fixed. In this case, an image domain is only allowed to be segmented into two parts in each layer. For an SAR image composed of N regions, the proceeding of the hierarchical strategy in each iteration is described as follows.

Assuming that every point in an SAR image is associated with a fuzzy membership family $\{u_1(x), u_2(x), \ldots, u_N(x)\}$ that belongs to the convex set [0,1] and is subjected to the two constraint conditions in (7). Each region is denoted by S_i $(i = 1, 2, \ldots, N)$ and satisfies $\mathrm{S}_i \bigcap \mathrm{S}_j = \varnothing$ $(i \neq j)$, $\bigcup_{i=1}^{N} \mathrm{S}_i = \Omega$, where Ω is the image domain. In the first layer, Ω is segmented into Ω_1 and $\bar{\Omega}_1$ by using $u_1(x)$ and fixing the rest of FMFs $u_i(x)$ $(i = 2, 3, \ldots, N)$, where $\Omega_1 + \bar{\Omega}_1 = \Omega$. We then take $\mathrm{S}_1 = \Omega_1$. In the second layer, the domain Ω is segmented into Ω_2 and $\bar{\Omega}_2$ by using $u_2(x)$ and fixing the rest of FMFs $u_i(x)$ $(i = 1, 3, \ldots, N, i \neq 2)$, where $\Omega_2 + \bar{\Omega}_2 = \Omega$. Please note that the segmentation of S_1 is confirmed in the first layer and only the subdomain $\bar{\Omega}_1$ is allowed to be segmented, we therefore take $\mathrm{S}_2 = \Omega_1 \bigcap \Omega_2$. Proceeding to the following layers, we can obtain $\mathrm{S}_k = \bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \bigcap \Omega_k$ $(k = 2, \ldots, N-1)$ and $\mathrm{S}_N = \bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \bigcap \bar{\Omega}_{N-1}$. Consequently, the segmentation is obtained by

$$\begin{aligned}\{\mathrm{S}_1, \mathrm{S}_2, \ldots, \mathrm{S}_{N-1}, \mathrm{S}_N\} &= \Big\{\Omega_1, \bar{\Omega}_1 \bigcap \Omega_2, \ldots, \bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \\ &\quad \bigcap \Omega_{N-1}, \bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \bigcap \bar{\Omega}_{N-1}\Big\}. \end{aligned} \tag{11}$$

For example, a four-multiphase segmentation in terms of the hierarchical strategy is shown in Fig. 4.

In the fuzzy region competition-based segmentation models [20], [28], to which region a point is classified is decided by its corresponding greatest FMF $u_i(x)$ as follows:

$$\{x \in \mathrm{S}_k | u_k(x) > u_i(x), i = 1, 2, \ldots, N, i \neq k\}. \tag{12}$$

In the proposed framework, the domain of an SAR image is only allowed to be partitioned into two parts Ω_i and $\bar{\Omega}_i$ $(i = 1, 2, \ldots, N-1)$ in each layer. The segmentation of the region in each layer is dependent on the segmentation result obtained from its upper layer. Therefore, in order to determine whether a point belongs to the subdomain Ω_i $(i = 1, 2, \ldots, N-1)$, we only need to compare two variables layer by layer. It is important to note that the fuzzy membership family belongs to the specified convex set [0,1]. We can, thereby, use the constraint condition (i) $\sum_{i=1}^{N} u_i(x) = 1$ in (7) here directly. Consequently, for the first layer, we obtain

$$\Omega_1(u_1) = \{x \in \Omega_1 | u_1(x) > 1 - u_1(x)\}. \tag{13}$$

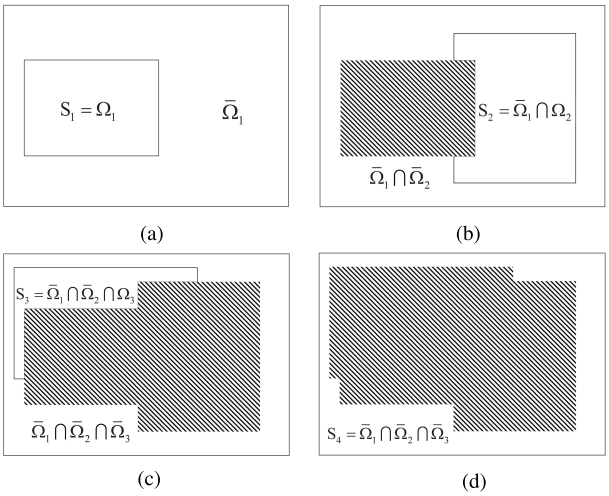


Fig. 4. Four-multiphase segmentation sketch based on the proposed framework. Segmentation on (a) the first layer, (b) the second layer, (c) the third layer, and (d) S_4.

For the second layer, we obtain

$$\Omega_2(u_1, u_2) = \{x \in \Omega_2 | u_2(x) > 1 - u_1(x) - u_2(x)\}. \tag{14}$$

Proceeding to the following layers, point x belonging to $\Omega_k(u_1, \ldots, u_k)$ $(k = 1, 2, \ldots, N-1)$ is decided by the formula as follows:

$$\Omega_k(u_1, \ldots, u_k) = \left\{ x \in \Omega_k | u_k(x) > \frac{1}{2}\left(1 - \sum_{i=1}^{k-1} u_i(x)\right)\right\}. \tag{15}$$

Combining (11) and (15), the multiphase segmentation of an SAR image based on the hierarchical strategy with respect to FMFs is achieved.

B. Energy Functional Formation and Its Convexity and Solutions

Based on the hierarchical strategy, the energy functional with respect to (4) can be rewritten as

$$\begin{aligned}\mathbf{E} &= \int_{\Omega_1} \boldsymbol{r}_1(x)\mathrm{d}x + \int_{\bar{\Omega}_1 \bigcap \Omega_2} \boldsymbol{r}_2(x)\mathrm{d}x + \cdots \\ &\quad + \int_{\bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \bigcap \Omega_{N-1}} \boldsymbol{r}_{N-1}(x)\mathrm{d}x \\ &\quad + \int_{\bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \bigcap \bar{\Omega}_{N-1}} \boldsymbol{r}_N(x)\mathrm{d}x + \sum_{i=1}^{N} \oint_{\partial\Omega_i} \mathrm{d}l \\ &= \sum_{i=1}^{N} \int_{\mathrm{R}_i} \boldsymbol{r}_i(x)\mathrm{d}x + \sum_{i=1}^{N} \oint_{\partial\Omega_i} \mathrm{d}l. \end{aligned} \tag{16}$$

In (16), $\boldsymbol{r}_i(x)$ measures the characteristics of an SAR image within i-region.

The minimization of (16) is implemented using an iteration method. In each iteration, it is calculated layer by layer using an FMF and fixing the rest of FMFs. To avoid an ambiguous derivation in each layer with respect to all FMFs, we use an indicator vector $\boldsymbol{\eta}(x) = [\eta^1(x), \eta^2(x), \ldots, \eta^N(x)]$ instead of direct using FMFs to indicate whether a point belongs to a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

certain subdomain, where $\eta^i(x)$ denotes $x \in \Omega_i$ and $\tilde{\eta}^i(x)$ denotes $x \notin \Omega_i$. In Section III-A, $u_i(x)$ corresponds to Ω_i, and $u_j(x)$ $(j = 1, 2, \ldots, N, i \neq j)$ corresponds to $\bar{\Omega}_i$. Using the constraint conditions (i) in (7), $\bar{\Omega}_i$ can be represented by the formula $\sum_{j=1, j\neq i}^{N} u_j(x) = 1 - u_i(x)$. We, thereby, simply define $\eta^i(x) = u_i(x)$ and $\tilde{\eta}^i(x) = 1 - u_i(x)$. For the two-phase segmentation, (16) degenerates to Mory's model presented in [20], and thus, we only consider the case $N \geq 3$.

Starting with the first layer, we can rewrite (16) as

$$\mathbf{E}[\{\Omega_1\}] = \int_{\Omega_1} \boldsymbol{r}_1(x)\mathrm{d}x + \int_{\bar{\Omega}_1} \boldsymbol{R}_1(x)\mathrm{d}x + \sum_{i=1}^{N} \oint_{\partial\Omega_i} \mathrm{d}l \quad (17)$$

where $\boldsymbol{R}_1(x) = \boldsymbol{r}_2(x)\eta^2(x) + \boldsymbol{r}_3(x)\tilde{\eta}^2(x)\eta^3(x) + \cdots + \boldsymbol{r}_{N-1}(x)\tilde{\eta}^2(x)\tilde{\eta}^3(x)\cdots\eta^{N-1}(x) + \boldsymbol{r}_N(x)\tilde{\eta}^2(x)\tilde{\eta}^3(x)\cdots\tilde{\eta}^{N-1}(x)$.

By associating FMF $u_1(x)$ with (17), subdomain Ω_1 is then extended to the whole domain in the integration. Following the same line of the work in relevant to the fuzzy region competition-based models [20], [21], [28], the functional for the first layer yields:

$$\begin{aligned}\mathbf{E}[\{\Omega_1\}] &= \int_{\Omega} \boldsymbol{r}_1(x)u_1(x)\mathrm{d}x + \int_{\Omega} \boldsymbol{R}_1(x)(1-u_1(x))\mathrm{d}x \\ &\quad + \int_{\Omega} |\nabla u_1(x)|\mathrm{d}x + \sum_{i=2}^{N} \oint_{\partial\Omega_i} \mathrm{d}l \\ &= \int_{\Omega} (\boldsymbol{r}_1(x) - \boldsymbol{R}_1(x))u_1(x)\mathrm{d}x + \int_{\Omega} |\nabla u_1(x)|\mathrm{d}x. \quad (18)\end{aligned}$$

In (18), the two terms $\int_{\Omega} \boldsymbol{R}_1(x)\mathrm{d}x$ and $\sum_{i=2}^{N} \oint_{\partial\Omega_i} \mathrm{d}l$ obtained in the process of algebraic operation have no dependence on $u_1(x)$. In the process of minimizing (18) using $u_1(x)$, the values of these two terms are irrelevant to the segmentation result in the first layer. These two terms are thus removed in the following calculation in this layer.

It is important to note that the formation of the simplified formula in (18) is the same as that of Mory's model in (5). In other words, the calculation of (18) on the first layer is referring to the special case of the two-phase fuzzy region competition-based segmentation model. Reference [20] has proved that this model with respect to a bounded and convex FMF set is convex. Since the used $u_1(x)$ in (18) belongs to the bounded and convex set [0,1], $\mathbf{E}[\{\Omega_1\}]$ is positive and convex and, thereby, has only globally optimal solutions. We refer readers to the literature [20] for the detailed proof of this convexity.

Based on the convex optimization theory, we employ Chambolle's dual-projection algorithm to attain a fast minimization of (18) [23], which is exactly as already being applied in [20] and [28]. In this fast global optimization algorithm, an auxiliary variable $v(x)$ is introduced in the calculation. Consequently, (18) is rewritten as

$$\begin{aligned}\mathbf{E}[\{\Omega_1\}] = \int_{\Omega} (\boldsymbol{r}_1(x) - \boldsymbol{R}_1(x))u_1(x)\mathrm{d}x + \int_{\Omega} |\nabla v_1(x)|\mathrm{d}x \\ + \frac{1}{2\theta}\int_{\Omega} (v_1(x) - u_1(x))^2\mathrm{d}x \quad (19)\end{aligned}$$

where θ is the weighting parameter for $v_1(x)$ and $u_1(x)$.

The minimizer of (19) is attained by minimizing (19) with respect to $v_1(x)$ and $u_1(x)$ separately. The auxiliary function $v_1(x)$ can be computed by fixing $u_1(x)$ and vice versa. After removing the inessential terms, we first need to minimize a functional with respect to $v_1(x)$ as follows:

$$\mathbf{E}(v_1(x)) = \int_{\Omega} |\nabla v_1(x)|\mathrm{d}x + \frac{1}{2\theta}\int_{\Omega} (v_1(x) - u_1(x))^2\mathrm{d}x. \quad (20)$$

Following the same line of [23] and [24], the solution of this functional obtained by the dual-projection algorithm is as follows:

$$v_1(x) = u_1(x) - \theta \mathrm{div} p_1^{n+1}(x) \quad (21)$$

where $p_1^{n+1}(x) = (p_1^n(x) + \tau\nabla(\mathrm{div}p_1^n(x) - u_1(x)/\theta))/(1 + \tau|\nabla(\mathrm{div}p_1^n(x) - u_1(x)/\theta)|)$, div is divergence, τ is the step size, and n is the current iterative number.

Then, by fixing $v_1(x)$, $u_1(x)$ is obtained by minimizing a functional as follows:

$$\begin{aligned}\mathbf{E}(u_1(x)) = \int_{\Omega} (\boldsymbol{r}_1(x) - \boldsymbol{R}_1(x))u_1(x)\mathrm{d}x \\ + \frac{1}{2\theta}\int_{\Omega} (v_1(x) - u_1(x))^2\mathrm{d}x. \quad (22)\end{aligned}$$

The solution of $u_1(x)$ obtained by the formula $\partial\mathbf{E}(u_1(x))/\partial u_1(x) = 0$ is given by

$$u_1(x) = v_1(x) - \theta(\boldsymbol{r}_1(x) - \boldsymbol{R}_1(x)). \quad (23)$$

To satisfy the constraint condition (ii) in (7), we choose to follow [28] to obtain an approximate solution of $u_1(x)$. According to (8), the corresponding approximate solution $\hat{u}_1(x)$ is obtained by

$$\hat{u}_1(x) = \min\{\max\{u_1(x), 0\}, 1\}. \quad (24)$$

The segmentation on the second layer relies on the segmentation result of the first layer. Equation (16) is thus rewritten with respect to the second layer as follows:

$$\begin{aligned}&\mathbf{E}[\{\Omega_2\}] \\ &= \int_{\Omega_1} \boldsymbol{r}_1(x)\mathrm{d}x + \int_{\bar{\Omega}_1 \bigcap \Omega_2} \boldsymbol{r}_2(x)\mathrm{d}x \\ &\quad + \int_{\bar{\Omega}_1 \bigcap \bar{\Omega}_2} \boldsymbol{R}_2(x)\mathrm{d}x + \oint_{\partial\Omega_2} \mathrm{d}l + \sum_{i=1, i\neq 2}^{N} \oint_{\partial\Omega_i} \mathrm{d}l \quad (25)\end{aligned}$$

where $\boldsymbol{R}_2(x) = \boldsymbol{r}_3(x)\eta^3(x) + \boldsymbol{r}_4(x)\tilde{\eta}^3(x)\eta^4(x) + \cdots + \boldsymbol{r}_{N-1}(x)\tilde{\eta}^3(x)\tilde{\eta}^4(x)\cdots\eta^{N-1}(x) + \boldsymbol{r}_N(x)\tilde{\eta}^3(x)\tilde{\eta}^4(x)\cdots\tilde{\eta}^{N-1}(x)$.

It is noted that $1 - u_1(x)$ corresponds to $\bar{\Omega}_1$ and $1 - u_2(x)$ corresponds to $\bar{\Omega}_2$. By associating the FMFs with (25),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

the energy functional for the second layer yields

$$\begin{aligned}\mathbf{E}[\{\Omega_2\}] &= \int_\Omega \boldsymbol{r}_1(x)u_1(x)\mathrm{d}x + \int_\Omega \boldsymbol{r}_2(x)(1-u_1(x))u_2(x)\mathrm{d}x \\ &\quad + \int_\Omega \boldsymbol{R}_2(x)(1-u_1(x))(1-u_2(x))\mathrm{d}x \\ &\quad + \int_\Omega |\nabla u_2(x)|\mathrm{d}x + \sum_{i=1,i\neq 2}^{N} \oint_{\partial\Omega_i} \mathrm{d}l \\ &= \int_\Omega [(\boldsymbol{r}_2(x)-\boldsymbol{R}_2(x))(1-u_1(x))]u_2(x)\mathrm{d}x \\ &\quad + \int_\Omega |\nabla u_2(x)|\mathrm{d}x. \end{aligned} \tag{26}$$

In (26), the segmentation on the first layer using $u_1(x)$ is achieved. For the second layer, the segmentation is implemented by using only $u_2(x)$, and other FMFs are treated as constants. The two terms $\int_\Omega[\boldsymbol{r}_1(x)u_1(x)+\boldsymbol{R}_2(x)(1-u_1(x))]\mathrm{d}x$ and $\sum_{i=1,i\neq 2}^{N}\oint_{\partial\Omega_i}\mathrm{d}l$ obtained in the process of algebraic operation that have no connection with $u_2(x)$ are, thereby, not taken into account in the following calculation for this layer.

It is interesting to observe that the formation of (26) with respect to $u_2(x)$ also pertains to the case of the two-phase fuzzy region competition-based segmentation model. Since $u_2(x)$ belongs to the bounded and convex set [0,1], based on the proof in [20], $\mathbf{E}[\{\Omega_2\}]$ is positive and convex. To minimize (26), we then apply Chambolle's dual-projection algorithm again, which is exactly the same of the ones applied in (18) [23]. Consequently, the solution for $v_2(x)$ is obtained by substituting $u_2(x)$ for $u_1(x)$ into (21), and the solution for $u_2(x)$ is $u_2(x) = v_2(x) - \theta(1-u_1(x))(\boldsymbol{r}_2(x)-\boldsymbol{R}_2(x))$.

To ensure the convexity of the energy functional, $u_2(x)$ also needs to satisfy the constraint conditions in (7). In the proposed framework, because $\hat{u}_1(x)$ has been determined in the segmentation of the first layer, the approximate values of $u_2(x)$ depend on that of $\hat{u}_1(x)$. Here, we use the constraint condition (i) $\sum_{i=1}^{N} u_i(x) = 1$, and we can easily obtain the approximate solution $\hat{u}_2(x)$ as follows:

$$\hat{u}_2(x) = \min\{\max\{u_2(x),0\}, 1-\hat{u}_1(x)\}. \tag{27}$$

Proceeding for the kth layer ($k = 1, 2, \ldots, N-1$), it is found that the energy functional minimized at each layer is all referring to the special case of the two-phase fuzzy region competition-based segmentation model. Therefore, Chambolle's dual-projection algorithm can be used repeatedly for obtaining the minimizers at each layer. The solutions for $v_k(x)$ and $u_k(x)$ are as follows:

$$\begin{aligned} v_k(x) &= u_k(x) - \theta \mathrm{div}\, p_k^{n+1}(x) \\ u_k(x) &= v_k(x) - \theta\psi(x,k)(\boldsymbol{r}_k(x)-\boldsymbol{R}_k(x)) \end{aligned} \tag{28}$$

where

$$\psi(x,k) = \begin{cases} 1, & k=1 \\ \prod_{i=1}^{k-1}(1-u_i(x)), & k>1 \end{cases}$$

and

$$\boldsymbol{R}_k(x) = \begin{cases} \boldsymbol{r}_{k+1}(x)\eta^{k+1}(x) + \boldsymbol{r}_{k+2}(x)\tilde{\eta}^{k+1}(x)\eta^{k+2}(x) \\ \quad +\cdots+\boldsymbol{r}_{N-1}(x)\tilde{\eta}^{k+1}(x)\tilde{\eta}^{k+2}(x)\cdots\eta^{N-1}(x) & k<N-1 \\ \quad +\boldsymbol{r}_N(x)\tilde{\eta}^{k+1}(x)\tilde{\eta}^{k+2}(x)\cdots\tilde{\eta}^{N-1}(x), \\ \boldsymbol{r}_N(x)\eta^N(x), & k=N-1. \end{cases}$$

The approximate solutions $\hat{u}_k(x)$ ($k = 1, 2, \ldots, N-1$) are

$$\hat{u}_k(x) = \min\left\{\max\{u_k(x),0\}, 1-\sum_{i=1}^{k-1}\hat{u}_i(x)\right\}. \tag{29}$$

Finally, $\hat{u}_N(x)$ is obtained by $\hat{u}_N(x) = 1 - \sum_{i=1}^{N-1}\hat{u}_i(x)$.

The energy functional associated with each FMF for every layer is verified to be positive and convex according to [20], and the corresponding solution for every layer is obtained. The last thing is to prove that the proposed framework related to the energy functional in (16) associated with all FMFs is convex, and the union of the solution of $u_i(x)$ with respect to each layer is the global minimizers of this framework.

In the proposed framework, the energy functional $\mathbf{E}$ with respect to FMFs is written in terms of the hierarchical form as follows:

$$\begin{aligned}&\mathbf{E}[\Omega; u_i(x)|_{i=1}^N] \\ &\quad = \mathbf{E}[\Omega_1; u_1(x)] + \mathbf{E}\left[\bar{\Omega}_1 \bigcap \Omega_2; u_2(x)\right] \\ &\qquad + \cdots + \mathbf{E}\left[\bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \bigcap \Omega_{N-1}; u_{N-1}(x)\right] \\ &\qquad + \mathbf{E}\left[\bar{\Omega}_1 \bigcap \bar{\Omega}_2 \bigcap \cdots \bigcap \bar{\Omega}_{N-1}; u_N(x)\right]. \end{aligned} \tag{30}$$

For simplicity, we refer to each term in (30), e.g., $\mathbf{E}[\bar{\Omega}_1 \bigcap \Omega_2; u_2(x)]$, as subfunctional. The domain corresponded to each subfunctional is mutually disjoint, and the union of them covers the image domain. In the proposed framework, because the calculation on the layer corresponded to a certain subdomain will not affect the segmentation result obtained from its upper layers, the calculation on such subdomain can be extended to that on the whole image domain. Based on this, minimizing (30) is exactly equivalent to the minimization of the formula as follows:

$$\begin{aligned}\mathbf{E}[\Omega; u_i(x)|_{i=1}^N] &= \mathbf{E}[\Omega_1; u_1(x)] + \mathbf{E}[\Omega_2; u_2(x)] \\ &\quad + \cdots + \mathbf{E}[\Omega_N; u_N(x)]. \end{aligned} \tag{31}$$

In (31), each subfunctional is the energy functional corresponded to every layer as reported previously. Every subfunctional is the special case of the two-phase fuzzy region competition-based segmentation model. By using (29), each subfunctional associated with a bounded and convex FMF is positive and convex [20]. It is well known that a nonnegative linear combination of convex functions is also convex. Hence, (31) with a bounded convex set is positive and convex. (29) ensures that all FMFs satisfy the constraint conditions $\hat{u}_i(x) \in [0,1]$ and $\sum_{i=1}^{N}\hat{u}_i(x) = 1$ in each iteration during the calculation, namely they belong to the convex set [0,1]. Equation (30) is consequently able to converge unconditionally to the global solutions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

According to the numerical calculating theory, $u_i(x)$ in (31) can be solved by fixing $u_j(x)$ $(j = 1, 2, \ldots, N, j \neq i)$ using the iteration method. The solution of $u_i(x)$ with respect to (31) is obtained by

$$u_i(x) = f(\mathbf{E}[\Omega_i; u_i(x)]). \tag{32}$$

It is obvious that the solution of $u_i(x)$ is exactly the same as (28) obtained at the ith layer. Therefore, the union of the solution of $u_i(x)$ associated with each layer is the global optimal solutions of the energy functional in (30).

C. Estimation of Characteristics Parameters and the Termination Criterion

The variables related to the error functions $\boldsymbol{r}_i(x)$ within each region S_i need to be estimated in each iteration during the calculation. Similarly, based on the numerical calculating theory, the estimation of the variables in $\boldsymbol{r}_i(x)$ can be obtained with $u_i(x)$ and $v_i(x)$ fixed. Here, we choose the CV model as an example [11]. In the CV model, the error function is written as

$$\boldsymbol{r}_i(x) = (I(x) - c_i(x))^2, \quad i = 1, 2, \ldots, N \tag{33}$$

where $I(x)$ denotes the image intensity and $c_i(x)$ is the statistical mean values within S_i and needs to be estimated in each iteration.

The energy functional in (16) with respect to the CV model is as follows:

$$\mathbf{E} = \sum_{i=1}^{N} \int_{\mathrm{R}_i} (I(x) - c_i(x))^2 \, \mathrm{d}x + \sum_{i=1}^{N} \oint_{\partial\Omega_i} \mathrm{d}l. \tag{34}$$

$c_i(x)$ is estimated layer by layer by fixing $u_i(x)$ and $v_i(x)$. After removing the constant terms irrelevant to $c_i(x)$ during the calculation, the estimated $c_i(x)$ is obtained by

$$\hat{c}_i(x) = \frac{\int_\Omega I(x)\psi(x,i)u_i(x)}{\int_\Omega \psi(x,i)u_i(x)}. \tag{35}$$

Other various variables in error functions can be estimated in the same way.

The termination criterion is defined in terms of the small time overhead principle as follows:

$$\int_\Omega \sum_{i=1}^{N} \left(u_i^{n+1}(x) - u_i^n(x)\right)^2 \mathrm{d}x < \mathrm{eps} \times \mathrm{num} \tag{36}$$

where eps is the control variable, n is the iterative number, and num is the total number of the pixels.

To avoid an undesirable program stop caused by a little change between two iterations, the program keeps running unless (36) is satisfied in continuous ten iterations. Consequently, the flowchart/pseudocode for the proposed segmentation framework is outlined as follows.

1) Initialize FMFs randomly in terms of (7). Set iterative number n.
2) Initialize auxiliary functions as $v_i(x) = 0$ and $p_i(x) = 0$ $(i = 1, 2, \ldots, N)$.
3) Start with $i = 1$. Minimize the energy functional in (16) layer by layer.
4) Estimate variables in error functions $\boldsymbol{r}_i(x)$.
5) Update $v_i(x)$ and $u_i(x)$ using (28); then, obtain the appropriate solution $\hat{u}_i(x)$ using (29).
6) If $i = N-1$, go to the next step; otherwise, set $i = i+1$ and go back to 4.
7) Obtain $\hat{u}_N(x) = 1 - \sum_{i=1}^{N-1} \hat{u}_i(x)$.
8) Check the termination criterion, if it is met in continuous ten iterations, then go to the next step; otherwise, go back to 3.
9) Achieve image segmentation by (11) and (15).

IV. Experimental Results and Discussion

Some parameters in the proposed framework need to be set before the segmentation. In the period of iterative numerical calculations, the weighting parameter θ in the dual=projection algorithm is set as 1 for a symmetry between FMFs $u_i(x)$ and auxiliary functions $v_i(x)$. In terms of iterative step τ, a large value of it improves the convergence rate while reducing the stability of the algorithm; we, thereby, set $\tau = 1/8$. Besides, we set $eps = 10^{-3}$, which is applicable to most of the images. For the two-phase segmentation, the proposed framework degenerates to Mory's model presented in [20]. Only multiphase segmentation ($N \geq 3$) is therefore taken into account in the experiments.

The experiments fall into two parts. The first part is to investigate the segmentation performances of the SAR images between the multiphase fuzzy region competition-based framework and the proposed framework. The second part is to test the efficiency and the performance of the proposed framework against the other two representative methods, the MRF-based segmentation framework and the level set-based framework. The MRF approach has been largely used in the SAR image segmentation, which is mainly developed based on the idea of the literature [35]. The level set is another most commonly used approach in the SAR image segmentation, which has been extensively studied with the variational methods and partial differential equations based on the minimization of the energy functional. The following experiments are carried out with MATLAB R2017a on Windows 10. The computing platform is with Intel Core i5-8400 CPU and 16 GB of memory.

A. Segmentation Performances Between the Multiphase Fuzzy Region Competition-Based Model and the Proposed Framework

In this part, we will apply the proposed framework to the fully polarimetric SAR (PolSAR) image described in Section II-C, as shown in Fig. 1. Section II-C has indicated that the application of the multiphase fuzzy region competition-based model to this SAR image cannot attain a correct segmentation result, as shown in Fig. 3. For an unbiased comparison, the error functions $\boldsymbol{r}_i(x)$ used in the proposed framework are defined in terms of the complex Wishart distribution as well, which is exactly the same as that in the multiphase fuzzy region competition-based model. The specific expressions of $\boldsymbol{r}_i(x)$ are given in (9) and (10). The image is segmented into three regions.

Fig. 5 shows the number of segmented pixels within each region in every iteration. It is observed that the three curves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

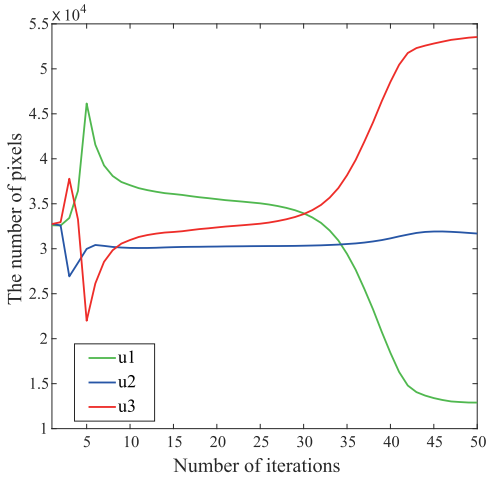


Fig. 5. Number of pixels within each segmented region with respect to Fig. 1 using (16).

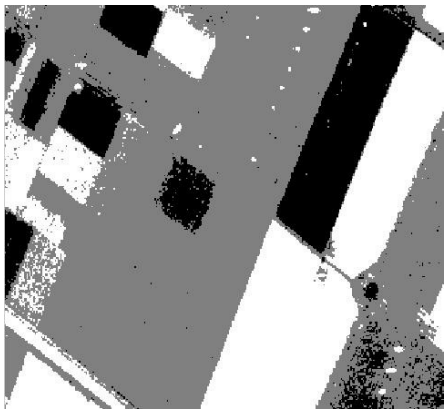


Fig. 6. Segmentation result by the proposed framework.

change smoothly after several iterations and remain unchanged eventually. It verifies the convergence of the solutions of the proposed framework. Its segmentation result is shown in Fig. 6. Based on the visual examination from Fig. 1, it is observed that Fig. 6 exhibits a much more better segmentation performance against Fig. 3. This validates the superiority of the proposed framework compared with the fuzzy region competition-based framework.

It should be noted that our proposed framework provides a general framework for the segmentation of SAR images. For a better segmentation performance, other useful information can be associated with the error functions $\boldsymbol{r}_i(x)$ in (16) in terms of different SAR images. Many related studies have shown that the effective ways to enhance the segmentation performance include the employment of edge information and textual information, the utilization of appropriate statistical distributions for certain regions, and the introduction of prior information [19], [36], [37]. The methods of improving segmentation performance for a certain SAR image are not the focus of this article. The proposed framework aims to offer a useful and effective tool for future research in the related field [38].

B. Comparisons Against the MRF-Based Framework and the Level Set-Based Framework

The MRF-based segmentation framework, the level set-based segmentation framework, and the proposed segmentation framework are all achieved by maximizing *a posteriori* probability under the Bayesian theory. The implementation of these three frameworks relies on the characteristics of SAR images. Generally, the statistical properties are the most commonly used information in the segmentation of images. To unbiasedly test the performances of the frameworks, the used statistical properties are exactly the same in the three frameworks. The segmentation accuracies are evaluated by four evaluation parameters, including user accuracy (UA), producer accuracy (PA), overall accuracy (OA), and kappa coefficient. A higher kappa coefficient implies a better segmentation performance.

The first tested image is a simulated SAR image composed of three objects with different shapes: a triangle (T), a star (S), and an annulus (A) and background (B). This simulated SAR image is of 360 × 360 pixels and is polluted by speckle noise with zero mean and 0.05 variance, as shown in Fig. 7(a). The SAR data obtained from the single channel are usually considered to follow the gamma distribution expressed as follows:

$$p(I(x)|\Omega_i) = \frac{1}{\Gamma(L)}\left(\frac{L}{c_i(x)}\right)^L I(x)^{L-1}\exp\left(-\frac{LI(x)}{c_i(x)}\right) \quad (37)$$

where Γ is the gamma function, L is the number of looks, $I(x)$ is the image intensity, and $c_i(x)$ is the mean value within i-region.

The gamma distribution in (37) is used in the MRF-based segmentation framework and is associated with the error functions $\boldsymbol{r}_i(x)$ in the level set-based framework and the proposed framework for the segmentation of this simulated SAR image. The segmentation results are shown in Fig. 7(b)–(d), and their corresponding accuracies and running time are shown in Table I.

We can see from Fig. 7 that the three shapes and the background are segmented separately with imprecise boundaries by the three frameworks. The real edges between two different shapes are still contaminated by noise. This can be attributed to the fact that the used gamma distribution is only suited to homogenous or low heterogeneous areas of intensity SAR images [39]. It is probably less capable of fitting the data of this simulated SAR image in comparison with other statistical distributions [40]. In addition, the three methods rely on the regional information to determine the segmentation boundaries, whereas some literature shows that the models using only regional information perform badly in the segmentation of different regions that are of the similar statistical properties [41]. It is noted once again that this experiment aims to investigate the segmentation performances of the three frameworks. The methods to improve the segmentation accuracy with respect to different SAR images, which have been discussed in Section IV-A, are not the focus of this experiment.

Table I shows that the proposed framework attains the best OA of 95.17% and the kappa coefficient of 0.915,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

TABLE I

SEGMENTATION ACCURACIES ATTAINED BY THE THREE FRAMEWORKS WITH RESPECT TO FIG. 7 AND THE CORRESPONDING RUNNING TIME

	The MRF-based framework				The level set-based framework				The proposed framework			
	T	S	A	B	T	S	A	B	T	S	A	B
UA (%)	89.68	81.72	71.09	95.80	87.92	92.17	68.76	98.19	91.63	97.48	83.69	97.51
PA (%)	91.46	86.55	88.58	90.29	92.84	88.47	89.75	93.24	96.90	91.53	94.81	95.82
OA (%)	89.71				92.13				95.17			
Kappa	0.823				0.864				0.915			
Running time (s)	57.8				39.3				8.7			

* T, S, A, B stand for Triangle, Star, Annulus, and Background, respectively.

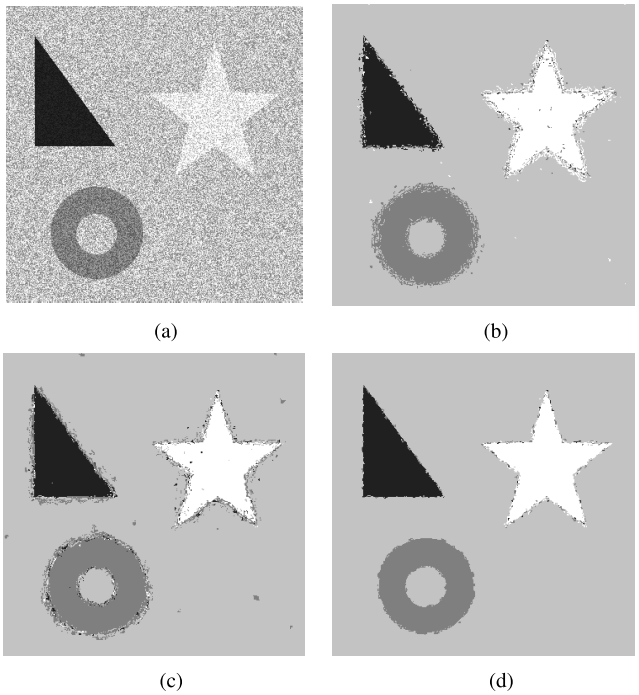


Fig. 7. Segmentation of the simulated SAR image by using the gamma model. (a) Tested simulated SAR image. Segmentation result by the (b) MRF-based framework, (c) level set-based framework, and (d) proposed framework.

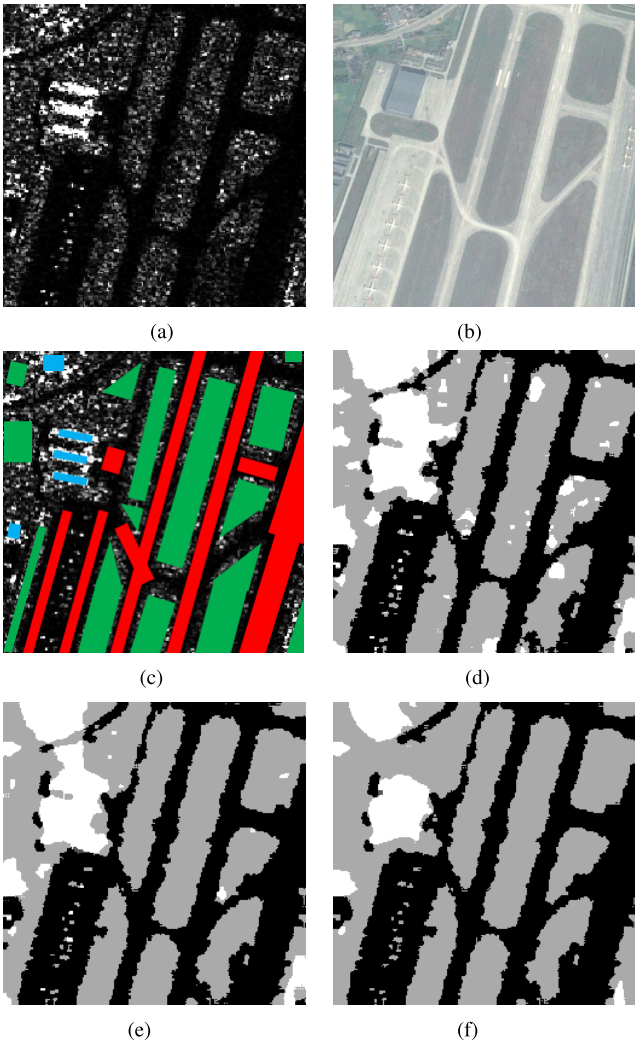


Fig. 8. Segmentation of the real SAR image by using G_0 model. (a) Tested real SAR image. (b) Corresponding optical image. (c) Validation data. (d) Segmentation result by the MRF-based framework. (e) Segmentation result by the level set-based framework. (f) Segmentation result by the proposed framework.

compared with the MRF-based framework and the level set-based framework, which exhibit the OA of 89.71% and the kappa coefficient of 0.823 and the OA of 92.13% of and the kappa coefficient of 0.864, respectively. The results indicate that in the case of the utilization of the exact same image characteristics, the proposed framework can achieve a better segmentation performance against the other two frameworks. Furthermore, the results indicate the high efficiency of the proposed framework. The running time of the proposed framework is 8.7 s, while that of the MRF-based framework and the level set-based framework is 57.8 and 39.3 s, respectively. The MRF-based framework involves the calculation of the spatial neighborhood information of every point in each iteration, which is time-consuming. The solving method for the level set-based framework is based on the variational principle and the gradient descent flow theorem. This locally convex solving method requires reinitialization of level set to distance functions in each iteration and is computationally inefficient. The proposed framework is minimized using the fast global optimization algorithm based on the convex optimization theory, which is highly efficient.

The second tested image is a real SAR image of 250 × 250 pixels as shown in Fig. 8(a), which was acquired from Radarsat-2 (C-band) on July 29, 2012 with HH polarization at 5.2 m × 7.7 m resolution. The image is mainly composed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SEGMENTATION ACCURACIES ATTAINED BY THE THREE FRAMEWORKS WITH RESPECT TO FIG. 8 AND THE CORRESPONDING RUNNING TIME

	The MRF-based framework			The level set-based framework			The proposed framework		
	R	L	C	R	L	C	R	L	C
UA (%)	93.44	98.54	17.70	98.57	97.93	44.22	96.51	98.87	36.98
PA (%)	97.77	73.66	93.72	97.84	84.19	94.06	97.95	89.93	94.71
OA (%)		84.14			89.89			93.36	
Kappa		0.729			0.827			0.877	
Running time (s)		86.4			51.3			10.7	

* R, L, S stand for Runways, Lawn, and Constructions, respectively.

of three regions, including lawn (L), construction (C), and runways (R). The corresponding optical images acquired on January 23, 2013 from the Google Earth as shown in Fig. 8(b) are used to obtain the validation data. The selected validation data set obtained by the visual examination with respect to the runways, the lawn, and the constructions is denoted by red, green, and blue, respectively, as shown in Fig. 8(c). In this experiment, the data of SAR images are assumed to be completely described by the G_0 distribution. The G_0 distribution can model the SAR data precisely for heterogeneous and extremely heterogeneous regions and is widely used in many applications [42]. The formation of G_0 distribution is as follows:

$$p(I(x)|\Omega_i) = \frac{L^L \Gamma(L-\alpha_i) I(x)^{L-1}}{\gamma_i^{\alpha_i} \Gamma(L)(\gamma_i + LI(x))^{L-\alpha_i}} \tag{38}$$

where Γ is the gamma function, L is the number of looks, $I(x)$ is the image intensity, α_i is the shape parameter, and γ_i is the scale parameter. The last two parameters can be estimated in terms of high-order moment with respect to i-region. We refer readers to [42] for the detailed information. Similarly, the G_0 distribution is used in the MRF-based segmentation framework and is associated with the error functions $\boldsymbol{r}_i(x)$ in the level set-based framework and the proposed framework for the segmentation of this real SAR image.

The segmentation results and the corresponding segmentation accuracies and the running time attained by the three frameworks are shown in Fig. 8(d)–(f) and Table II, respectively. Fig. 8(f) clearly shows that the proposed framework achieves the best segmentation performance with the smooth and correct boundaries compared with Figs. 8(d) and 8(e). Table II also demonstrates that the proposed framework attains a higher OA of 93.36% and a kappa coefficient of 0.877 compared with the MRF-based framework and the level set-based framework, which exhibit an OA of 84.14% and a kappa coefficient of 0.729 and an OA of 89.89% and a kappa coefficient of 0.827, respectively. For instance, based on the visual analysis, the construction at the top-left image is separated with correct boundaries by the proposed framework, whereas its adjacent pixels that do not belong to such region are misclassified as "constructions" by the MRF-based framework and the level set-based framework.

The running time of the MRF-based framework and the level set-based framework for this SAR image is 86.4 and 51.3 s, respectively, while the proposed framework consumes only 10.7 s, which verifies the high efficiency of the proposed framework.

The third tested image is a fully PolSAR image of 325 × 325 pixels, which was acquired by AIRSAR at L-band from Altona, Canada, with a resolution of 5 m × 5 m. The corresponding backscattering total power image is shown in Fig. 9(a). The observed area is mainly composed of three crops. Since the corresponding optical image acquired at the same period as that of the SAR image is not accessible at the moment, instead, the validation data are given based on the visual examination, as shown in Fig. 9(b). In Fig. 9(b), the different regions are denoted by red (R), blue (B), and green (G), respectively.

To implement the segmentation, for simplicity, each point of the PolSAR image is assumed to exactly follow the complex Wishart distribution. Therefore, the complex Wishart distribution in (9) is used in the MRF-based framework and is linked with error functions $\boldsymbol{r}_i(x)$ in the level set-based framework and the proposed framework. The segmentation results attained by the three frameworks are shown in Fig. 9(c)–(e), and their segmentation accuracies and the running time are shown in Table III.

Fig. 9 clearly shows that the proposed framework attains the best performance compared with the other two frameworks. In Fig. 9(c), the segmentation boundaries are not smooth, and some small regions are misclassified from the homogenous regions. The level set-based framework performs better compared with the MRF-based framework. However, some areas still are classified into the wrong class. For instance, the region in the bottom-right corner of the image is falsely separated into two parts. The proposed framework achieves the expected boundaries with statistically homogenous regions. Table III quantitatively verifies the efficiency and effectiveness of the proposed method. In comparison with the MRF-based framework and the level set-based framework, the OA of the proposed framework increases by 12.54% and 4.96%, respectively, and the kappa coefficient of that increases by 0.193 and 0.076, respectively. The proposed framework also attains much higher efficiency with 14.8 s, against that with 69.1 and 102.2 s, respectively.

The experiments on the simulated SAR image, the real SAR image acquired from single channel, and the fully PolSAR image indicate that by using the exact same image properties, the proposed framework can attain a better segmentation performance and much higher efficient against the MRF-based framework and the level set-based framework. The proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

TABLE III
SEGMENTATION ACCURACIES ATTAINED BY THE THREE FRAMEWORKS WITH RESPECT TO FIG. 9 AND THE CORRESPONDING RUNNING TIME

	The MRF-based framework			The level set-based framework			The proposed framework		
	G	R	B	G	R	B	G	R	B
UA (%)	92.48	89.96	56.40	98.30	93.07	72.93	98.78	97.09	85.76
PA (%)	77.01	95.77	67.99	86.27	97.77	83.58	94.16	96.94	94.04
OA (%)		82.71			90.29			95.25	
Kappa		0.733			0.850			0.926	
Running time (s)		102.2			69.1			14.8	

* G, R, B stand for the regions corresponded to Green, Red, and Blue in Fig. 9(b).

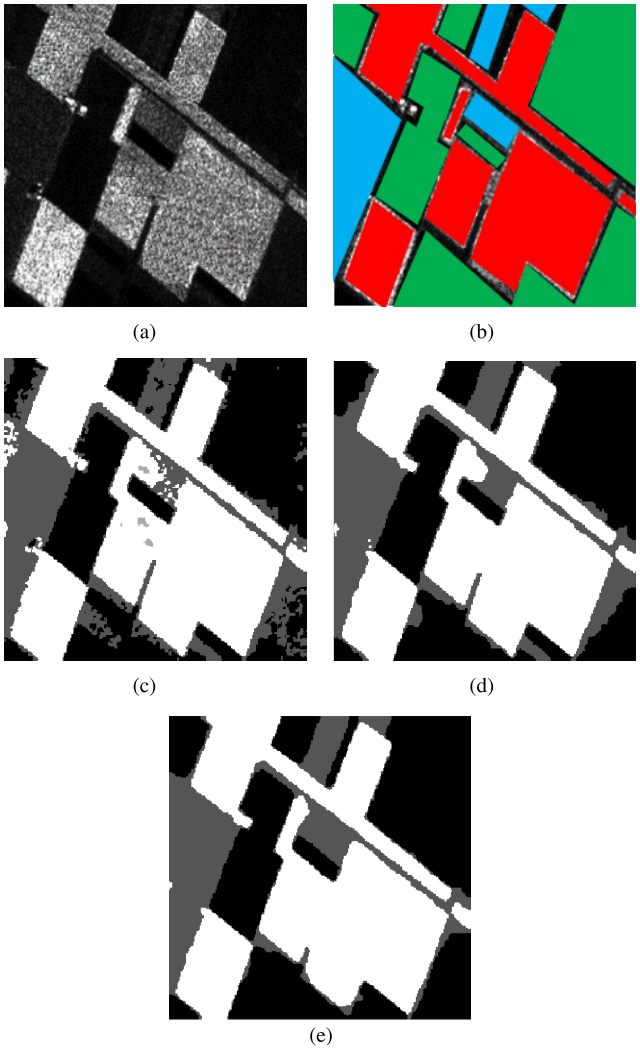


Fig. 9. Segmentation of the fully PolSAR image by using a complex Wishart distribution model. (a) Tested fully PolSAR image. (b) Validation data. Segmentation result by the (c) MRF-based framework, (d) level set-based framework, and (e) proposed framework.

framework contributes to offering a useful and effective tool in the segmentation of SAR images for the future study.

V. CONCLUSION

Based on the idea of the two-phase fuzzy region competition-based segmentation method [20], we propose an improved multiphase segmentation framework for SAR images in terms of the hierarchical strategy. The proposed framework partitions an SAR image layer by layer. In each layer, the subdomain corresponded to the image, which is determined by the segmentation results of its upper layers, is segmented into only two parts. The proposed framework not only inherits the advantages of the fuzzy region competition-based segmentation method but also is suited to the segmentation of images corrupted with strong noise. The experimental results show that by using the same image characteristics, the proposed framework can attain a better segmentation performance and much highly efficient compared with the commonly used segmentation frameworks: the MRF-based framework and the level set-based framework. Furthermore, the proposed framework can be generalized to various types of SAR images composed of multiple regions. By using multiple features as well as by associating appropriate characteristics with error functions in the involved energy functional for a certain SAR image, the proposed framework is expected to achieve a satisfactory segmentation performance. The proposed framework offers a useful and efficient segmentation tool for future studies.

REFERENCES

[1] X. Yang, X. Gao, D. Tao, X. Li, and J. Li, "An efficient MRF embedded level set method for image segmentation," *IEEE Trans. Image Process.*, vol. 24, no. 1, pp. 9–21, Jan. 2015.
[2] H. M. Moftah, A. T. Azar, E. T. Al-Shammari, N. I. Ghali, A. E. Hassanien, and M. Shoman, "Adaptive k-means clustering algorithm for MR breast image segmentation," *Neural Comput. Appl.*, vol. 24, nos. 7–8, pp. 1917–1928, 2014.
[3] A. N. Benaichouche, H. Oulhadj, and P. Siarry, "Multiobjective improved spatial fuzzy C-means clustering for image segmentation combining Pareto-optimal clusters," *J. Heuristics*, vol. 22, no. 4, pp. 383–404, 2016.
[4] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Region-based convolutional networks for accurate object detection and segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 1, pp. 142–158, Jan. 2016.
[5] S. Guolei and H. Wei, "Overview of SAR image segmentation algorithm," *Comput. Era*, vol. 5, pp. 1–5, May 2017.
[6] M. Kass, A. Witkin, and D. Terzopoulos, "Snakes: Active contour models," *Int. J. Comput. Vis.*, vol. 1, no. 4, pp. 321–331, 1988.
[7] L. D. Cohen, "On active contour models and balloons," *CVGIP, Image Understand.*, vol. 53, no. 2, pp. 211–218, 1991.
[8] C. Xu and J. L. Prince, "Snakes, shapes, and gradient vector flow," *IEEE Trans. Image Process.*, vol. 7, no. 3, pp. 359–369, Sep. 1998.
[9] S. Osher and J. A. Sethian, "Fronts propagating with curvature-dependent speed: Algorithms based on Hamilton–Jacobi formulations," *J. Comput. Phys.*, vol. 79, no. 1, pp. 12–49, 1988.
[10] V. Caselles, R. Kimmel, and G. Sapiro, "Geodesic active contours," *Int. J. Comput. Vis.*, vol. 22, no. 1, pp. 61–79, 1997.
[11] T. F. Chan and L. A. Vese, "Active contours without edges," *IEEE Trans. Image Process.*, vol. 10, no. 2, pp. 266–277, Feb. 2001.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

[12] S. C. Zhu and A. Yuille, "Region competition: Unifying snakes, region growing, and Bayes/MDL for multiband image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 9, pp. 884–900, Sep. 1996.
[13] C. Samson, L. Blanc-Féraud, G. Aubert, and J. Zerubia, "A level set model for image classification," *Int. J. Comput. Vis.*, vol. 40, no. 3, pp. 187–197, 2000.
[14] C. Li, C. Xu, C. Gui, and M. D. Fox, "Level set evolution without re-initialization: A new variational formulation," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 430–436.
[15] M. Horritt, "A statistical active contour model for SAR image segmentation," *Image Vis. Comput.*, vol. 17, nos. 3–4, pp. 213–224, 1999.
[16] I. B. Ayed, A. Mitiche, and Z. Belhadj, "Multiregion level-set partitioning of synthetic aperture radar images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 5, pp. 793–800, May 2005.
[17] Y. Shuai, H. Sun, and G. Xu, "SAR image segmentation based on level set with stationary global minimum," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 4, pp. 644–648, Oct. 2008.
[18] I. B. Ayed, N. Hennane, and A. Mitiche, "Unsupervised variational image segmentation/classification using a Weibull observation model," *IEEE Trans. Image Process.*, vol. 15, no. 11, pp. 3431–3439, Nov. 2006.
[19] F. Galland, J.-M. Nicolas, H. Sportouche, M. Roche, F. Tupin, and P. Réfrégier, "Unsupervised synthetic aperture radar image segmentation using Fisher distributions," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 8, pp. 2966–2972, Aug. 2009.
[20] B. Mory and R. Ardon, "Fuzzy region competition: A convex two-phase segmentation framework," in *Proc. Int. Conf. Scale Space Variational Methods Comput. Vis.* Heidelberg, Germany: Springer, 2007, pp. 214–226.
[21] B. Mory, R. Ardon, and J.-P. Thiran, "Variational segmentation using fuzzy region competition and local non-parametric probability density functions," in *Proc. IEEE 11th Int. Conf. Comput. Vis.*, Oct. 2007, pp. 1–8.
[22] C. Li, R. Huang, Z. Ding, C. Gatenby, D. Metaxas, and J. Gore, "A variational level set approach to segmentation and bias correction of images with intensity inhomogeneity," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Heidelberg, Germany: Springer, 2008, pp. 1083–1091.
[23] A. Chambolle, "An algorithm for total variation minimization and applications," *J. Math. Imag. Vis.*, vol. 20, no. 1, pp. 89–97, 2004.
[24] X. Bresson, S. Esedoğlu, P. Vandergheynst, J.-P. Thiran, and S. Osher, "Fast global minimization of the active contour/snake model," *J. Math. Imag. Vis.*, vol. 28, no. 2, pp. 151–167, Jun. 2007.
[25] L. A. Vese and T. F. Chan, "A multiphase level set framework for image segmentation using the Mumford and Shah model," *Int. J. Comput. Vis.*, vol. 50, no. 3, pp. 271–293, Dec. 2002.
[26] T. Brox and J. Weickert, "Level set segmentation with multiple regions," *IEEE Trans. Image Process.*, vol. 15, no. 10, pp. 3213–3218, Oct. 2006.
[27] I. B. Ayed, A. Mitiche, and Z. Belhadj, "Polarimetric image segmentation via maximum-likelihood approximation and efficient multiphase level-sets," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 9, pp. 1493–1500, Sep. 2006.
[28] F. Li, M. K. Ng, T. Y. Zeng, and C. Shen, "A multiphase image segmentation method based on fuzzy region competition," *SIAM J. Imag. Sci.*, vol. 3, no. 3, pp. 277–299, 2010.
[29] Y. N. Law, H. K. Lee, and A. M. Yip, "A multiresolution stochastic level set method for Mumford–Shah image segmentation," *IEEE Trans. Image Process.*, vol. 17, no. 12, pp. 2289–2300, Nov. 2008.
[30] R. C. P. Marques, F. N. Medeiros, and J. S. Nobre, "SAR image segmentation based on level set approach and $\mathcal{G}_A^0$ model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 10, pp. 2046–2057, Dec. 2011.
[31] S. Y. Luo, L. Tong, Y. Chen, and M. Q. Jia, "SAR image segmentation based on the advanced level set," in *Proc. IOP Conf. Ser., Earth Environ. Sci.*, 2014, vol. 17, no. 1, Art. no. 012246.
[32] N. R. Pal and S. K. Pal, "A review on image segmentation techniques," *Pattern Recognit.*, vol. 26, no. 9, pp. 1277–1294, 1993.
[33] S. Y. Luo, L. Tong, and Y. Chen, "A multi-region segmentation method for SAR images based on the multi-texture model with level sets," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2560–2574, May 2018.
[34] J. S. Lee, M. R. Grunes, and R. Kwok, "Classification of multi-look polarimetric SAR imagery based on complex Wishart distribution," *Int. J. Remote Sens.*, vol. 15, no. 11, pp. 2299–2311, Jul. 1994.
[35] E. Rignot and R. Chellappa, "Segmentation of polarimetric synthetic aperture radar data," *IEEE Trans. Image Process.*, vol. 1, no. 3, pp. 281–300, Jul. 1992.
[36] Z. Cao, Y. Pi, X. Yang, and J. Xiong, "A variational level set SAR image segmentation approach based on statistical model," in *Proc. 7th Eur. Conf. Synth. Aperture Radar*, Jun. 2008, pp. 1–4.
[37] J. Yin and J. Yang, "A modified level set approach for segmentation of multiband polarimetric SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7222–7232, Nov. 2014.
[38] S. Luo, K. Sarabandi, L. Tong, and S. Guo, "Unsupervised multi-region partitioning of fully polarimetric SAR images with advanced fuzzy active contours," *IEEE Trans. Geosci. Remote Sens.*, to be published, doi: 10.1109/TGRS.2019.2947376.
[39] A. C. Frery, H.-J. Müller, C. C. F. Yanasse, and S. J. S. Sant'Anna, "A model for extremely heterogeneous clutter," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 3, pp. 648–659, May 1997.
[40] G. Gao, "Statistical modeling of SAR images: A survey," *Sensors*, vol. 10, no. 1, pp. 775–795, 2010.
[41] P. Yu, A. K. Qin, and D. A. Clausi, "Unsupervised polarimetric SAR image segmentation and classification using region growing with edge penalty," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1302–1317, Apr. 2012.
[42] C. C. Freitas, A. C. Frery, and A. H. Correia, "The polarimetric g distribution for SAR data analysis," *Environmetrics*, vol. 16, no. 1, pp. 13–31, 2005.



Shiyu Luo (S'16–M'19) received the B.S. degree in technique and instrumentation of measurements and the Ph.D. degree in instrument science and technology from the School of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2010 and 2017, respectively.

He was a Visiting Scholar from 2014 to 2016 and a Research Assistant from 2016 to 2018 with the Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, MI, USA. Since 2019, he has been a Post-Doctoral Fellow with the School of Automation Engineering, University of Electronic Science and Technology of China. His research interests include synthetic aperture radar data processing, synthetic aperture radar image segmentation and classification, and information retrieval from SAR images.



Kamal Sarabandi (S'87–M'90–SM'92–F'00) received the B.S. degree in electrical engineering from the Sharif University of Technology, Tehran, Iran, in 1980, and the M.S. degree in electrical engineering, the M.S. degree in mathematics, and the Ph.D. degree in electrical engineering from the University of Michigan, Ann Arbor, MI, USA, in 1986, 1989, and 1989, respectively.

He is currently the Director of the Radiation Laboratory, University of Michigan, where he was the Rufus S. Teesdale Endowed Professor of engineering with the Department of Electrical Engineering and Computer Science. He led the Center for Microelectronics and Sensors, University of Michigan, sponsored by the Army Research Laboratory under the Micro-Autonomous Systems and Technology Collaborative Technology Alliance Program from 2008 to 2018. He is also leading the Microwave Sensor Technology Center, University of Michigan. He has authored many book chapters and over 280 articles in refereed journals on miniaturized and on-chip antennas, metamaterials, electromagnetic scattering, wireless channel modeling, random media modeling, microwave measurement techniques, radar calibration, inverse scattering problems, and microwave sensors. He has also had more than 650 articles and invited presentations in many national and international conferences and symposia on similar subjects. His research interests include microwave- and millimeter-wave radar remote sensing, metamaterials, electromagnetic wave propagation, and antenna miniaturization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.

Dr. Sarabandi served as a member of the NASA Advisory Council appointed by the NASA Administrator for two consecutive terms from 2006 to 2010. He served as the President for the IEEE Geoscience and Remote Sensing Society (GRSS) in 2015 and 2016. He is a member of the Commissions F and B of the International Union of Radio Science (URSI). He was a recipient of the Henry Russel Award from the Regent of The University of Michigan in 1997, the GAAC Distinguished Lecturer Award from the German Federal Ministry for Education, Science, and Technology in 1999, the 1996 EECS Department Teaching Excellence Award, the IEEE GRSS Distinguished Achievement Award and the University of Michigan Faculty Recognition Award in 2005, the Best Article Award at the 2006 Army Science Conference, the IEEE GRSS Symposium Best Article Award in 2008 and 2017, the Humboldt Research Award from The Alexander von Humboldt Foundation of Germany in 2008, the 2010 Distinguished Faculty Achievement Award from the University of Michigan, the 2011 IEEE Judith A. Resnik Award from the IEEE Board of Directors, the IEEE GRSS with its 2013 Education Award, the College of Engineering Research Excellence Award in 2004, the Stephen S. Attwood Award in 2017, the Ted Kennedy Family Faculty Team Excellence Award in 2018 from the College of Engineering, University of Michigan, and the NASA Group Achievement Award for his contributions to the NASA SMAP Mission. He is serving as the Chair for the U.S. National Committee (USNC) URSI Commission F. In 2016, he was recognized as one of the top 50 graduates of the Sharif University of Technology. In the past several years, joint articles presented by his students at a number of international symposia, such as the 1995 IEEE Antennas and Propagation Society International Symposium (APS'95), APS'97, APS'00, APS'01, APS'03, APS'05, APS'06, APS'07, and APS'16, the 1999 IEEE International Geoscience and Remote Sensing Symposium (IGARSS'99), IGARSS'02, IGARSS'07, IGARSS'11, and IGARSS'14, the 2001 IEEE MTT-S International Microwave Symposium (IMS'01), USNC URSI'04, URSI'05, URSI'06, URSI'10, and URSI'11, AMTA'06, URSI GA'08 and GA'14, and the Eastern Snow Conference'16, have received the best article awards. He was a member of the Editorial Board of the PROCEEDINGS OF THE IEEE and an Associate Editor of the IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION and the IEEE SENSORS JOURNAL.



Ling Tong (M'06) received the B.S. and M.S. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1985 and 1988, respectively.

From 1988 to 1995, she was with the National Institute of Measurement and Testing Technology, Chengdu. She is currently the Director of the Microwave Remote Sensing and Testing Laboratory, University of Electronic Science and Technology of China, where she is a Professor (second grade) with the School of Automatic Engineering. She is also leading the Earth Observation Engineering Research Center of Sichuan Province, UESTC. Her research interests include the mechanism of remote sensing, microwave remote sensing, data analysis and processing, and metrical information theory.



Sen Guo received the B.S. degree in economics, the M.S. degree in industrial engineering, and the Ph.D. degree in energy management from North China Electric Power University, Beijing, China, in 2009, 2013, and 2016, respectively.

He is currently an Associate Professor and a Master Supervisor with the School of Economics and Management, North China Electric Power University. His research interests include energy economics and environment, energy price, and energy system optimization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:06:45 UTC from IEEE Xplore. Restrictions apply.